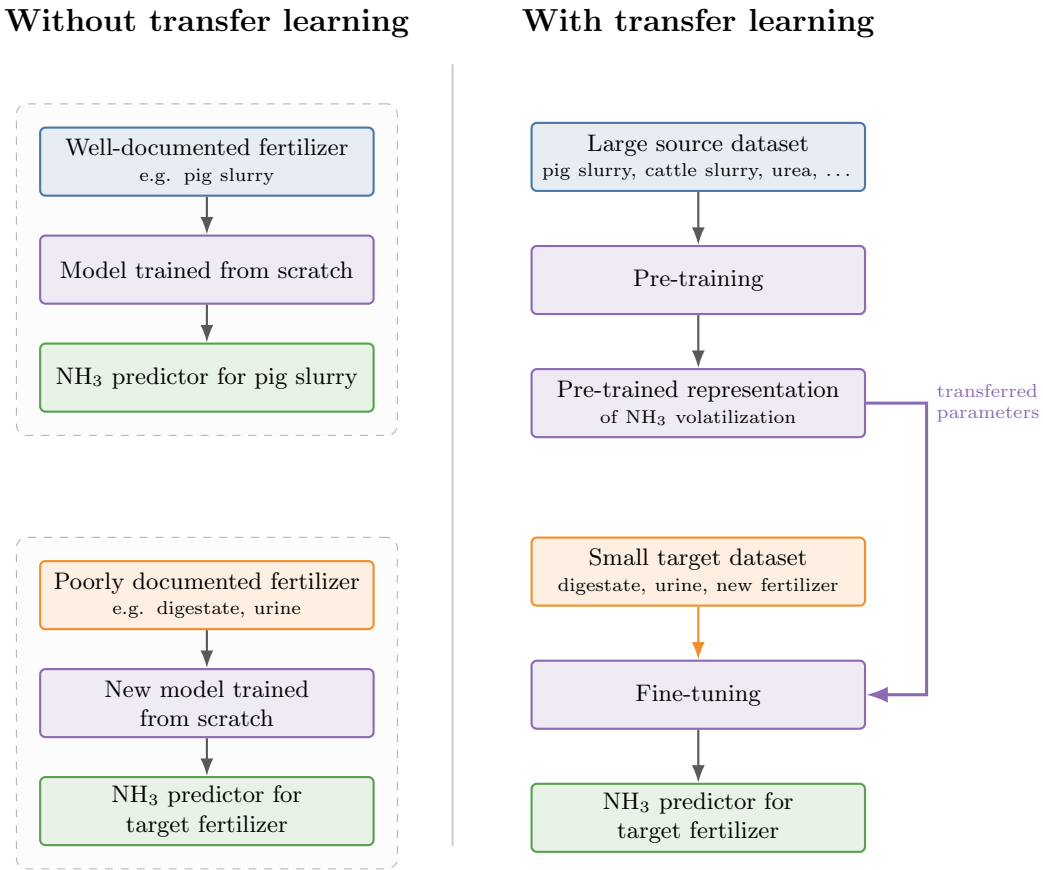

Without transfer learning
With transfer learning
Well-documented fertilizer
e.g. pig slurry
Model trained from scratch
NH3 predictor for pig slurry
Poorly documented fertilizer
e.g. digestate, urine
New model trained
from scratch
NH3 predictor for
target fertilizer
Large source dataset
pig slurry, cattle slurry, urea, ...
Pre-training
Pre-trained representation
of NH3 volatilization
transferred
parameters
Small target dataset
digestate, urine, new fertilizer
Fine-tuning
NH3 predictor for
target fertilizer